Project Report

Cooling Load Calculation of a Single-Zone Conditioned Space by the CLTD/CLF Method

Prepared by

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1. Introduction and Objective

The cooling load is the rate at which heat must be removed from a space to maintain the required indoor temperature and humidity. This load is needed before the air-conditioning equipment and other main system components can be sized. Heat can enter the room through the roof, walls, and windows, from solar radiation through the glass, from outdoor air entering the room, and from internal sources such as lights, people, and equipment.

A wall or roof exposed to the sun does not pass all of the absorbed heat into the room at once. Some of the heat is stored in the construction and reaches the indoor space later. The Cooling Load Temperature Difference/Cooling Load Factor (CLTD/CLF) method accounts for this delay by using tabulated temperature differences and load factors instead of solving the full transient heat-transfer problem. This makes the method convenient for hand calculations and preliminary design work.

In this project, the CLTD/CLF method is used to calculate the cooling load of a single-zone room. The calculation includes heat transfer through shaded and sunlit surfaces, solar heat gain through the windows, sensible and latent loads from outdoor air, and internal heat gains. The calculated load is then used to estimate the required cooling capacity and is also compared with a simple floor-area rule of thumb.

2. Problem Description

The conditioned space is 10 m × 10 m with a height of 3 m. Its floor area is therefore 100 m^2 and its volume is 300 m^3. The indoor design condition is 25.5 °C and 50% relative humidity. The north wall is next to an unconditioned space at 30 °C. The east wall and east window face the outdoor open space but do not receive direct solar radiation at the design hour. The west and south walls are exposed to the sun, and the roof is exposed to the outdoor environment. Figure 1 shows the room layout and the surrounding conditions.

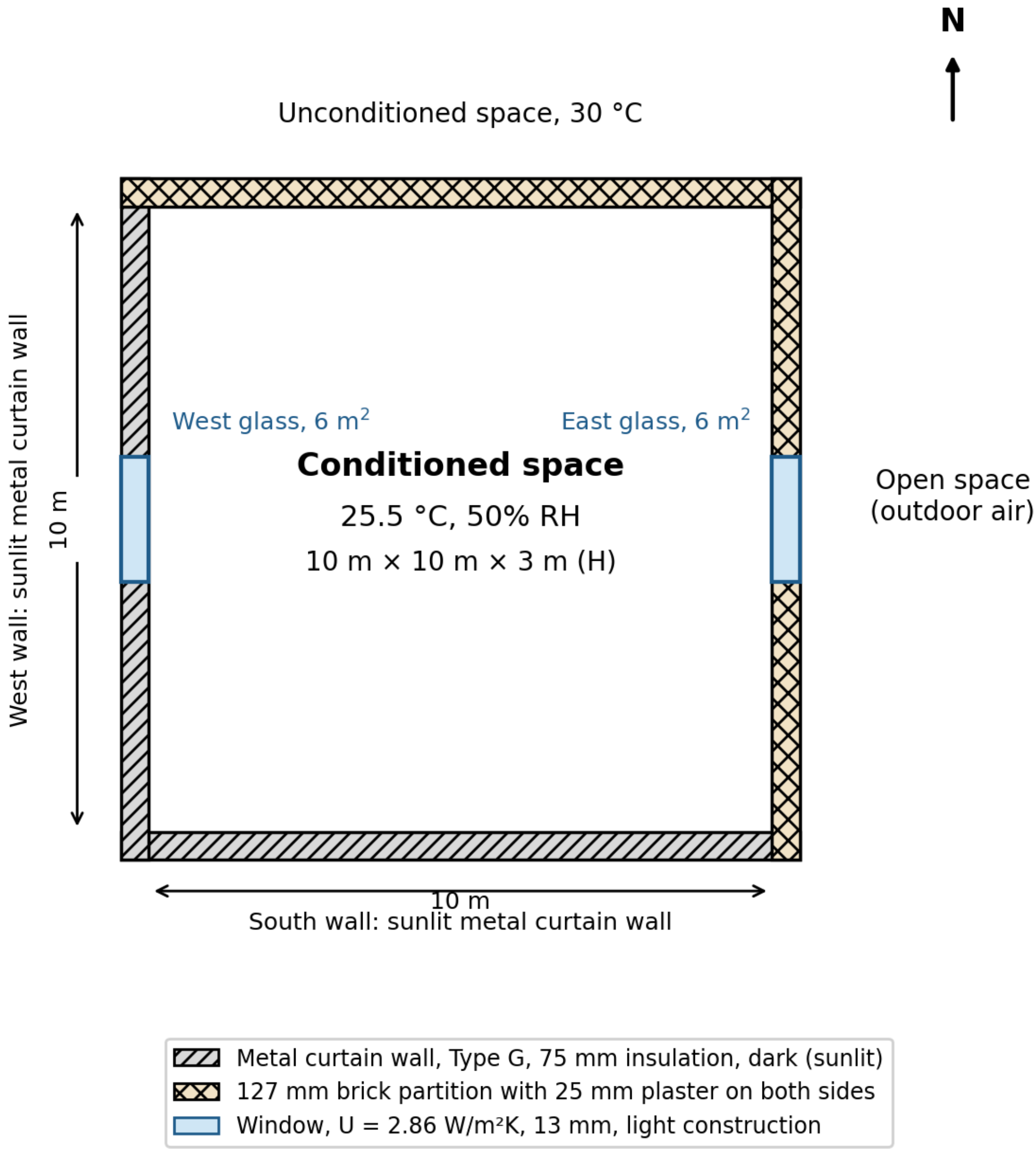

Figure 1: Plan of the conditioned space and its surroundings.

The construction details and design data used in the calculation are listed in Table 1. For this problem, the design calculation is carried out for 17:00 (5 PM) on 12 April. From the local weather data in Table 9 of the course text, the maximum outdoor dry-bulb temperature is 33 °C and the daily temperature range is 11 °C.

Item	Description
Roof	Type 4, 50 mm insulation, no suspended ceiling
West and south walls	Type G metal curtain wall, 75 mm insulation, sunlit, dark-colored
Brick walls (N, E)	127 mm brick with 25 mm plaster on both sides
Windows	6 m^2 each on the west and east sides; U = 2.86 W/m^2K (given), 13 mm thick, light construction
Indoor design condition	25.5 °C, 50% relative humidity

Item	Description
Surroundings	North: unconditioned space at 30 °C; East: open space (outdoor air)
Air exchange rate	1 air change per hour
Lighting	25 W/m^2 lighting power density, fluorescent, operated from 8:00 to 18:00
Occupancy	5 seated occupants, one computer in use, present from 8:00 to 18:00
Design hour	17:00, 12 April

Table 1: Construction details and design data.

Each wall has a gross area of 10 m × 3 m = 30 m^2. The west and east walls each contain a 6 m^2 window, so the remaining opaque wall area on each of those sides is 24 m^2. The south wall and the north wall each have an area of 30 m^2. The roof area is 100 m^2.

3. Method of Calculation

Heat transfer through the building envelope is handled differently for shaded and sunlit surfaces. For a shaded surface, the heat gain is calculated from the temperature difference across the surface:

$$Q = U \cdot A \cdot TD$$

where U is the overall heat transfer coefficient in W/m^2K, A is the surface area in m^2, and TD is the dry-bulb temperature difference across the surface.

For a sunlit wall or roof, solar heating and the thermal mass of the construction delay part of the heat gain. In the CLTD method, this effect is represented by a corrected cooling load temperature difference:

$$Q = U \cdot A \cdot CLTD_c$$

The tabulated CLTD values are based on reference conditions of 21 July, 40°N latitude, an indoor temperature of 25.5 °C, and an average outdoor temperature of 29.4 °C. Because the conditions in this problem are different, the tabulated CLTD values are corrected using:

$$CLTD_c = [\,(CLTD + LM)\,K + (25.5 - T_i) + (T_{o,av} - 29.4)\,]\,f$$

In this expression, LM is the latitude-month correction from Table 13, K is the color correction factor, Ti is the indoor design temperature, To,av is the average outdoor temperature for the design day, and f is the attic fan factor. For dark surfaces K = 1, while a value of 0.5 is used for light-colored surfaces. Here, f = 1 because no attic fan or duct ventilation is considered. Roof CLTD values are taken from Table 12(b), wall values from Table 15, and glass values from Table 20.

Solar heat gain through sunlit glass is calculated separately from the conductive heat gain. The solar contribution is found from:

$$Q = A \cdot SC \cdot SHGF_{max} \cdot CLF$$

where SC is the shading coefficient, SHGFmax is the maximum solar heat gain factor for the window orientation and month, and CLF is the cooling load factor. The CLF accounts for the delay between solar energy entering the room and the time when it appears as a cooling load.

Outdoor air adds both sensible and latent cooling loads. Heat from lighting, occupants, and equipment is treated as internal heat gain. These loads are calculated later in Sections 8 and 9.

4. Overall Heat Transfer Coefficients

The overall heat transfer coefficient, U, is calculated for each construction. Since the layers in these constructions are in series, their thermal resistances are added:

$$\Sigma R = R_o + R_1 + R_2 + \ldots + R_i\,, \quad U = 1 / \Sigma R$$

where Ro and Ri are the outside and inside surface film resistances. The wall and roof constructions used in this project are treated as simple series layers.

4.1 Roof

The roof is Type 4 with 50 mm of insulation and no suspended ceiling. Table 12(a) gives the layer codes as A0, E2, E3, B5, C12, and E0, and the corresponding resistance values are taken from Table 16. The B5 value in the table is for 25 mm of insulation. Since the problem specifies 50 mm, that resistance is doubled.

Code	Layer	R (m^2K/W)
A0	Outside surface resistance	0.059
E2	12 mm slag or stone	0.099
E3	10 mm felt and membrane	0.050
B5	Insulation, 25 mm tabulated, 50 mm used	0.587 × 2 = 1.174
C12	50 mm heavyweight concrete	0.029
E0	Inside surface resistance	0.121
	Total, ΣR	**1.532**

Table 2: Thermal resistances of the roof layers (series).

$$U_{roof} = 1 / 1.532 = 0.653\ W/m^2K$$

4.2 West and South Walls

The west and south walls are Type G metal curtain walls. From Table 14, the layer codes are A0, A3, B12, A3, and E0. B12 is used for the insulation layer because Table 16 identifies it as the 75 mm insulation specified in the problem.

Code	Layer	R (m^2K/W)
A0	Outside surface resistance	0.059
A3	Steel siding	0
B12	75 mm insulation	1.76
A3	Steel siding	0

Code	Layer	R (m^2K/W)
E0	Inside surface resistance	0.121
	Total, ΣR	**1.94**

Table 3: Thermal resistances of the west and south wall layers (series).

$$U_{wall} = 1 / 1.94 = 0.515\ W/m^2K$$

4.3 Brick Walls (North and East)

The north and east brick walls consist of 127 mm of brick with 25 mm of plaster on each side. Table 16 gives k = 0.727 W/mK for common brick. The resistance of the 127 mm brick layer is therefore calculated from its actual thickness:

$$R_{brick} = L / k = 0.127 / 0.727 = 0.175\ m^2K/W$$

For each 25 mm plaster layer, k = 0.7277 W/mK, giving a resistance of 0.0344 m^2K/W per side. The surface film resistances are taken from Table 10. The inside value is Ri = 0.12 m^2K/W for a non-reflective vertical surface. For the outside surface, the summer value Ro = 0.044 m^2K/W is used. Adding all of the resistances gives:

$$\Sigma R = 0.044 + 0.0344 + 0.175 + 0.0344 + 0.12 = 0.4078\ m^2K/W$$

$$U_{partition} = 1 / 0.4078 = 2.45\ W/m^2K$$

4.4 Glass

The window U-value is given directly as 2.86 W/m^2K, so no separate layer-by-layer resistance calculation is required for the glass.

5. Conduction Through Shaded Surfaces

Three surfaces are treated as shaded at the design hour. The north wall is next to the unconditioned space at 30 °C, so the temperature difference is 30 − 25.5 = 4.5 °C. The east brick wall and east window are exposed to outdoor air at 33 °C, giving a temperature difference of 33 − 25.5 = 7.5 °C. The calculated conduction loads are listed in Table 4.

Item	A (m^2)	U (W/m^2K)	TD (°C)	Q = U·A·TD (W)
North partition wall	30	2.45	4.5	330.75
East brick wall	24	2.45	7.5	441.00
East glass	6	2.86	7.5	128.70
Total				**900.45**

Table 4: Heat conduction through shaded walls and glass.

6. Conduction Through Sunlit Surfaces

6.1 Corrected Cooling Load Temperature Differences

The average outdoor temperature for the design day is needed before the CLTD correction can be applied. Using a maximum temperature of 33 °C and a daily range of 11 °C from Table 9:

$$T_{o,av} = T_{o,max} - DR / 2 = 33 - 11/2 = 27.5\ °C$$

The indoor temperature is the same as the CLTD reference value, Ti = 25.5 °C, so the indoor-temperature correction is zero. The sunlit surfaces are treated as dark, so K = 1. No attic fan or duct ventilation is considered, so f = 1.

Roof. For a Type 4 roof without a suspended ceiling, Table 12(b) gives CLTD = 37 °C at 17:00. The latitude-month correction for the horizontal roof is LM = 0. Therefore:

$$CLTD_c = (37 + 0)(1) + (25.5 - 25.5) + (27.5 - 29.4) = 35.1\ °C$$

West wall. Table 15 gives CLTD = 40 °C, and Table 13 gives LM = −0.5. Therefore:

$$CLTD_c = (40 - 0.5)(1) + 0 + (27.5 - 29.4) = 37.6\ °C$$

South wall. Table 15 gives CLTD = 17 °C, and Table 13 gives LM = −1.6. Therefore:

$$CLTD_c = (17 - 1.6)(1) + 0 + (27.5 - 29.4) = 13.5\ °C$$

West glass. For conductive heat gain through the west window, Table 20 gives a CLTD of 7 °C.

6.2 Conduction Loads

Item	A (m^2)	U (W/m^2K)	CLTDc (°C)	Q = U·A·CLTDc (W)
Roof (Type 4)	100	0.653	35.1	2292.03
South wall (Type G)	30	0.515	13.5	208.58
West wall (Type G)	24	0.515	37.6	464.74
West glass	6	2.86 (given)	7 (Table 20)	120.12
Total				3085.47

Table 5: Heat conduction through the roof and sunlit walls and glass.

7. Solar Radiation Through Glass

Heat gain through a window includes both conduction and solar radiation. At 17:00, the west window is exposed to the sun, while the east window is treated as shaded. For the west window, Table 21 gives SC = 0.88, Table 18(a) gives SHGFmax = 719 W/m² for April, and Table 19 gives CLF = 0.64. With a window area of 6 m²:

$$Q_{solar} = A \cdot SC \cdot SHGF_{max} \cdot CLF = 6 \times 0.88 \times 719 \times 0.64 = 2429.64\ W$$

The solar load through the west window is about 2.43 kW. This is slightly higher than the roof conduction load and much higher than the conduction load through any single wall or window. The result shows that the west-facing glass makes an important contribution to the cooling load in the late afternoon.

8. Cooling Load Due to Air Exchange

Outdoor air entering the room must be cooled from the outdoor condition to the indoor design condition, and moisture must also be removed. The sensible and latent loads are calculated from the outdoor-air flow rate as follows:

$$Q_s = \rho\, c_p\, \dot{V}\,(T_o - T_i) = 1200\, \dot{V}\,(T_o - T_i)$$

$$Q_L = \rho\, h_{fg}\, \dot{V}\,(w_o - w_i) = 3010 \times 10^3\, \dot{V}\,(w_o - w_i)$$

Here, ρ is the air density, cp is the specific heat of air, hfg is the latent heat of vaporization of water, $\dot{V}$ is the outdoor-air volume flow rate, and w is the humidity ratio. The simplified coefficients 1200 and 3010 × 10^3 used below are the values adopted for this calculation.

At one air change per hour, the 300 m^3 room requires an outdoor-air flow rate of $\dot{V}$ = 300/3600 = 0.0833 m^3/s. Using an outdoor temperature of 33 °C, the sensible load is:

$$Q_s = 1200 \times 0.0833 \times (33 - 25.5) = 750\ W$$

At 25.5 °C and 50% RH, the indoor humidity ratio is wi = 0.0102 kg/kg. The problem does not give an outdoor relative humidity, so 60% RH at 33 °C is assumed for this calculation. This gives wo = 0.0191 kg/kg. Since the latent load is sensitive to outdoor humidity, it should be recalculated if actual design humidity data are available. Using the assumed value:

$$Q_L = 3010 \times 10^3 \times 0.0833 \times (0.0191 - 0.0102) \approx 2232\ W$$

With this humidity assumption, the latent outdoor-air load is considerably larger than the sensible part. This shows that moisture removal can make up a significant part of the cooling requirement in warm and humid conditions.

9. Internal Heat Gains

Lighting. The lighting power density is 25 W/m² over a floor area of 100 m², giving an installed lighting load of 2500 W. A 25% ballast allowance is included for the fluorescent fixtures, so the sensible lighting gain is:

$$Q_{light} = 1.25 \times 25 \times 100 = 3125\ W$$

The lights are on from 8:00 to 18:00. In this calculation, the full lighting gain is used at 17:00 instead of applying a separate lighting CLF. This is a conservative simplification.

Occupants. Five seated occupants are considered. Sensible and latent heat gains of 70 W and 45 W per person, respectively, are used:

$$Q_{s,people} = 5 \times 70 = 350\ W\,, \quad Q_{L,people} = 5 \times 45 = 225\ W$$

Equipment. One computer is included with an assumed sensible heat gain of 150 W.

The occupant and computer heat-gain values are treated as design assumptions because the problem does not give separate values for them.

10. Load Summary and Equipment Selection

Table 6 brings together all of the calculated sensible and latent loads at the design hour.

Load component	Sensible (W)	Latent (W)	Total (W)
Conduction, shaded surfaces (Table 4)	900.45	–	900.45
Conduction, sunlit surfaces (Table 5)	3085.47	–	3085.47
Solar radiation, west glass	2429.64	–	2429.64
Air exchange (1 ACH)	750.00	2232.00	2982.00
Lighting (fluorescent, with ballast)	3125.00	–	3125.00
Occupants (5, seated)	350.00	225.00	575.00
Equipment (1 computer)	150.00	–	150.00
Total	10790.56	**2457.00**	13247.56

Table 6: Summary of the cooling load at 17:00, 12 April.

The total cooling load is about 13.25 kW. Using 1 TR = 3517 W, the load in tons of refrigeration is:

$$\text{Cooling load} = 13247.6 / 3517 = 3.77\ \text{TR}$$

A 10% design allowance is used in this project to account for uncertainty in weather conditions, air exchange, and future use. This increases the required capacity to about 14.6 kW, or 4.1 TR. A nominal 4.5 TR unit is therefore used as the project selection. For an actual installation, the final choice should be checked against the manufacturer's sensible and total capacity ratings at the design conditions.

For comparison, a simple floor-area rule of thumb uses about 1 TR for every 150 to 200 ft^2. The room area is 100 m^2, or about 1076 ft^2, which gives 5.4 to 7.2 TR by this rule. This is higher than the detailed calculation. The difference comes from the fact that the detailed method accounts for the actual wall and roof insulation, window area, occupancy, lighting, and outdoor-air load instead of using floor area alone.

11. Conclusion

The cooling load of the 10 m × 10 m × 3 m room was calculated with the CLTD/CLF method for 17:00 on 12 April. The shaded surfaces contribute about 900 W by conduction, while the sunlit roof, walls, and west glass contribute about 3085 W by conduction. Solar heat gain through the west window adds about 2430 W. The outdoor-air load is 750 W sensible and about 2232 W latent based on the assumed outdoor humidity. Lighting, occupants, and equipment add another 3625 W sensible and 225 W latent. The total cooling load is about 13.25 kW, or 3.77 TR. With the 10% design allowance used in this project, a nominal 4.5 TR capacity is selected.

The load breakdown also shows which parts of the room have the greatest effect on the result. Lighting is the largest single load listed in the summary. For the building envelope, the west window is an important contributor because it has both conductive and solar heat gain. Lower lighting power and better shading on the west window would reduce the calculated cooling load. The floor-area rule of thumb gives a noticeably larger capacity, showing why a detailed load calculation is more useful for equipment sizing.

References

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